OIT 367 - Business Intelligence from Big Data
Winter 2025
Final Project Report

ENHANCING DISCHARGE SUMMARY TRANSLATIONS WITH RAG

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EXECUTIVE SUMMARY

Problem

Language barriers in healthcare can lead to miscommunication and poor adherence to discharge instructions, especially for patients with Limited English Proficiency (LEP). Existing translation tools often fail to capture medical nuances, compromising patient safety; additionally, their lengthy processing times further delay essential care. This project explores using Retrieval-Augmented Generation (RAG) to augment large language models (LLMs) for more accurate Spanish translations of clinical discharge summaries. Using a dataset of clinician-verified translations from Stanford's STARR-OMOP database, we assess different LLMs, training-test splits, and few-shot learning configurations. Model performance is evaluated using BLEU scores and statistical analysis, with comparisons against Google Translate and the Helsinki NLP model to measure real-world improvements.

Principal Findings

The primary findings of our analysis were fourfold, attempting to evaluate the effectiveness of a medical-grade translation task by comparing average BLEU scores across: 1) different LLMs (GPT-3.5-Turbo, GPT 4, GPT o-1, Claude Sonnet 3.7), 2) varying random k-shot configurations (25, 50, 75, and 100-random shot), 3) varying RAG-based k-shot learning configurations (0, 5, 10 and 15-shot), and 4) existing standard-of-care models (Helsinki and Google Translate). For random few-shot, we determined that 50-shot was optimal. For RAG-based few-shot learning, we found that 5-shot was adequate, with 10-shot and 15-shot not showing significant gains over the 5-shot version. Overall, we found that the RAG-based 5-shot version of GPT-3.5-Turbo performed the best, with an average BLEU score of 0.471 out of 1.0 which was 20 percentage points higher than the untrained version (p-value = 4.72e-12). The trained models were all major improvements over traditional translation techniques like Helsinki (BLEU score = 0.246) and Google Translate (BLEU score = 0.245).

Recommendations

We recommend for rapid clinical documentation translation tasks that clinicians use GPT-3.5-Turbo with RAG-based 5-shot learning. Further, we recommend training on technically-relevant data (e.g., translated medical document summaries) rather than training on language data broadly. This recommendation is limited to Spanish language translation, as other languages were not evaluated in this study, and we advise a human-in-the-loop approach where a certified human translator is able to perform a final review prior to delivering potentially sensitive clinical information. In real-world settings, we note that current translation approaches including the use of Google Translate, Helsinki or untrained/zero-shot LLMs such as ChatGPT or Claude, should be advised against due to their inferior performance.

I. INTRODUCTION

Language barriers in healthcare significantly impact patient outcomes, particularly for individuals with Limited English Proficiency (LEP). Discharge summaries are a critical component of post-hospitalization care, providing patients with essential instructions regarding medications, follow-up visits, and potential complications. However, the lack of accurate and timely translations of these summaries into a patient's native language, particularly Spanish, poses challenges for effective communication and adherence to medical guidance. Stanford Medicine currently has a 48 hour wait time requirement on document translation requests. Interviews with primary care physicians indicate that patients often struggle to adhere to their discharge care plan due to a lack of available discharge instructions in their preferred language, leading to potential adverse health outcomes.

While existing tools like Google Translate and the Helsinki NLP model are commonly used, these often fail to capture the nuances of medical language, leading to inaccuracies that can compromise patient safety. This project explores the use of Retrieval-Augmented Generation (RAG) to train large language models (LLMs) for translating clinical discharge summaries into Spanish while maintaining medical accuracy and understandability.

There are two primary research questions addressed in this study:

1. How can RAG be leveraged to augment LLMs for improving the translation of clinical discharge summaries? More specifically, what is the optimal combination of LLM model, random few-shot, or RAG-based few-shot learning configuration to achieve the best translation performance?
2. How do translations produced by augmented LLMs compare with existing real-world solutions, such as Google Translate and the Helsinki NLP model?

By developing a machine-assisted translation system, we aim to support human translators, reduce translation workloads, and improve the overall quality of healthcare for Spanish-speaking patients.

II. DATA

The dataset used in this study originates from Stanford's STARR-OMOP database, which contains electronic health records from patients admitted to Stanford Hospital. Institutional approval for medical records analysis was obtained under Stanford IRB 74122. All identifying medical information was scraped in accordance with Stanford IRB and HIPAA compliance protocols.

The dataset consists of 20 discharge summaries from patients admitted between 2021 and 2022, further broken down into 194 phrases for more granular evaluation. Each phrase has been manually translated by Spanish-speaking clinicians, serving as a gold-standard reference for evaluating model performance.

The dataset is structured as a table with 194 rows, where each row represents a unique phrase in English and its corresponding Spanish translation. There are two primary columns:

1. Column 1: Original English discharge summary
2. Column 2: Clinician-conducted Spanish translation

See Appendix 5.0 for example of the dataset.

III. METHODS

Our approach involves three phases to assess the effectiveness of LLM-based translation models:

Phase 1 - Model Design

In this phase, we determined the optimal random few-shot and RAG-based few-shot learning configuration to maximize translation performance, using GPT-3.5-Turbo and GPT-4 as our primary comparison models. This involved:

1. **Shot size selection for random few-shot learning:** We experimented with random 25-shot, 50-shot, 75-shot, and 100-shot learning to determine the optimal number of examples needed for effective translation. The dataset of 194 phrases was divided into training (fed into prompt as part of few-shot) and test sets with varying ratios to determine the optimal split for preventing overfitting and underfitting. For instance, a random 25-shot corresponds to a test set of 169 phrases to calculate the BLEU score. Due to the limited dataset size, we opted not to maintain a fixed test set, though we acknowledge that doing so could have better isolated the impact of training set variations.
2. **Shot size selection for RAG-based few-shot learning:** We experimented with RAG-based zero-shot, 5-shot, 10-shot, and 15-shot learning to determine the optimal number of examples needed for effective translation. We believe few-shot learning is particularly relevant here given limited training data. We used OpenAI's embedding model (`text-embedding-3-small`) and cosine similarity selection to identify the most contextually similar examples from the training dataset, which were then fed into the prompt.

Phase 2 - Model Selection

We then evaluated four different LLMs, GPT-3.5-Turbo, GPT-4, GPT o-1 , Claude Sonnet 3.7, across three different model configurations:

1. **Baseline comparison with an untrained LLM**: This approach tests model performance without any few-shot learning, serving as a control condition.
2. **Random few-shot learning:** This model randomly selects k human-translated phrases to feed into the prompt as reference sentences. We will leverage the optimal random k-shot size found in Phase 1.
3. **Few-shot learning with RAG:** This model uses RAG to retrieve relevant human-translated phrases from the training set during translation to feed into the prompt as reference sentences. We will leverage the optimal RAG-based k-shot found in Phase 1.

Model outputs were evaluated using BiLingual Evaluation Understudy (BLEU) scores, which assess how closely the model-generated translations align with human translations on a scale of 0 to 1, with a score of 1 indicating a perfect translation match. We used box-and whisker plots to compare translation performance across models, and paired t-tests to assess statistical significance in translation accuracy among models.

Phase 3 - Comparison with Existing Translation Solutions

Finally, to validate real-world applicability, we benchmarked the best-performing model from Phase 2 against current methods, including:

1. Google Translate (widely used by the general population).
2. Helsinki NLP Translation Model (commonly used in medical translation).

These models were selected based on informational interviews with Stanford Medicine staff members who reported their use in clinical care. We evaluated the average BLEU score produced by Google Translate and Helsinki over the 194 phrases, enabling us to quantify the improvement of our LLMs over existing methods.

IV. RESULTS

Phase 1 - Model Design

1. **Shot size selection for random few-shot learning:**
   We evaluated the impact of different random few-shot learning configurations (25, 50, 75, and 100-shot) on translation accuracy. Our experiments showed that the optimal random few-shot size was 50-shot, with improved performance across all models. The untrained version of GPT-3.5-Turbo had an average BLEU score 0.273, which increased to 0.381 with random few-shot; likewise, the untrained version of GPT 4 had an average BLEU score of 0.256, which increased to 0.281 with random few-shot. As shown in Appendix 5.1, BLEU score improvement from increasing the random-shot size plateaued

around n=50, with decreasing performance past n=75 likely due to additional reference phrases not providing any additional value.

2. **Shot size selection for RAG-based few-shot learning:**
We evaluated the impact of different RAG-based few-shot learning configurations (0, 5, 10, and 15-shot) on translation accuracy. We found that 5-shot learning had the best performance gains across models, with an average BLEU score of 0.471 for GPT-3.5-Turbo and 0.338 for GPT-4. This was significantly improved over a zero shot approach (i.e. prompting LLMs without training), which had average BLEU scores of 0.273 for GPT-3.5-Turbo (Appendix 5.2) and 0.256 for GPT-4 (Appendix 5.3). The improvement seen by the 5-shot model over the 0-shot model was statistically significant using a paired t-test ($p = 1.48e-32$ for GPT 3.5 and $p = 1.67e-17$ for GPT-4). We found that model improvement plateaued beyond k=5, with the 10-shot and 15-shot models not showing meaningful increases in performance beyond the 5-shot model.

We ran a paired t-test on the BLEU scores for each of the 194 samples produced by the zero-shot and RAG-based 5-shot models in order to assess whether the RAG-based 5-shot version had a statistically significant improvement on translation performance. As shown in Exhibit 4.0, the RAG-based 5-shot version of every model (Claude, GPT-3.5-Turbo, GPT4, and GPT-o1) significantly outperformed their untrained counterparts, with p-values all below 0.000002. GPT-3.5-Turbo had the largest improvement, showing an average BLEU score increase of 20.4 percentage points (95% Confidence Interval: (+0.176, +0.232)). GPT4 came in second with a BLEU score improvement of 0.088, followed by Claude (+0.067) and GPTo1 (+0.064).

Exhibit 4.0: Paired t-tests comparing the BLEU scores produced by RAG-based 5-shot LLMs vs their untrained (0-shot) counterparts.

Model	**Average improvement** **(5-shot model vs 0-shot model)**	**Standard Error**	**T-statistic**	**P-value**	**95% Confidence Interval**
Claude	0.0674	0.0091	7.375	4.72e-12	(0.0494, 0.0854)
GPT-3.5-Turbo	0.2042	0.0141	14.450	1.48e-32	(0.1763, 0.232)
GPT 4	0.0883	0.0094	9.398	1.67e-17	(0.0697, 0.1068)
GPT o1	0.0641	0.0127	5.028	1.13e-06	(0.0389, 0.0892)

Phase 2 - Model Selection

We compared four different LLMs (GPT-3.5-Turbo, GPT-4, GPT-o1, Claude Sonnet 3.7) on the translation task varying across three model configurations: untrained (zero-shot), random 50-shot, and RAG-based 5-shot.

- The overall best performing model was the RAG-based 5-shot GPT-3.5-Turbo model, which had an average BLEU score of 0.471 (Std Error ±0.245) (highlighted in green in Exhibit 4.1)
- The best performing random 50-shot model was GPT-3.5-Turbo, which had a BLEU score of 0.381 (Std Error ±0.229) (highlighted in yellow in Exhibit 4.1)
- The best performing untrained model was GPT-o1, which had an average BLEU score of 0.310 (Std Error ±0.190) (highlighted in red in Exhibit 4.1)

Exhibit 4.1: BLEU score performance on untrained (zero-shot), random 50-shot, and 5-shot LLMs. * indicates best overall, ** indicates best random 50-shot, *** indicates best untrained

Model	**Untrained**	**Random 50-shot**	**RAG-based 5-shot**
Claude Sonnet 3.7	0.274 (0.183)	0.275 (0.194)	0.341 (0.200)
GPT-3.5-Turbo	0.273 (0.187)	0.381 (0.229)**	0.471 (0.245)*
GPT 4	0.256 (0.186)	0.281 (0.185)	0.338 (0.202)
GPT o1	0.310 (0.190)***	0.328 (0.196)	0.374 (0.209)

Additional figures comparing the models can be found in Appendix 5.4.

Phase 3 -Comparison with Existing Translation Solutions

In order to benchmark our results, we also evaluated the performance of the LLMs with existing translation tools: Google translate and Hugging-Face's Helsinki-NLP (Appendix 5.5). Both of these models had similar average performance, with BLEU scores for Google Translate of 0.2446 (SD = ±0.158) and Helsinki-NLP of 0.2457 (SD = ±0.1628). These scores are slightly below the untrained LLMs, which scored in the range of 0.26 - 0.31, and significantly worse than the random 50-shot or RAG-based 5-shot LLMs, which scored in the range of 0.34 - 0.47.

Exhibit 4.2: Sample translation comparisons between best performing trained model and current Standard of Care models.

English	**Clinician-Verified Spanish**	**GPT-3.5 Turbo (5-shot)**	**Google Translate**	**Helsinki NLP**
"Take 500 mg of	"Tome 500 mg de	"Tome 500 mg	"Tomar 500 mg	"Toma 500 mg

acetaminophen every 6 hours as needed for pain."	acetaminofén cada 6 horas según sea necesario para el dolor."	de acetaminofén cada 6 horas si necesita aliviar el dolor."	de acetaminofén cada seis horas para el dolor."	de paracetamol cada 6 horas cuando el dolor lo requiera."

In this sample sentence, which includes a simple discharge instruction with a sentence in subject-verb form, we note that Google Translate fails to capture conditional phrasing (eg. no mention of 'as needed') and Helsinki introduces a new word choice (using paracetamol rather than acetaminophen). While these inconsistencies may result in the same clinical outcome for patients in this example, this can be unsafe if clinical text is being mistranslated at scale and for more complex queries.

V. CONCLUSION

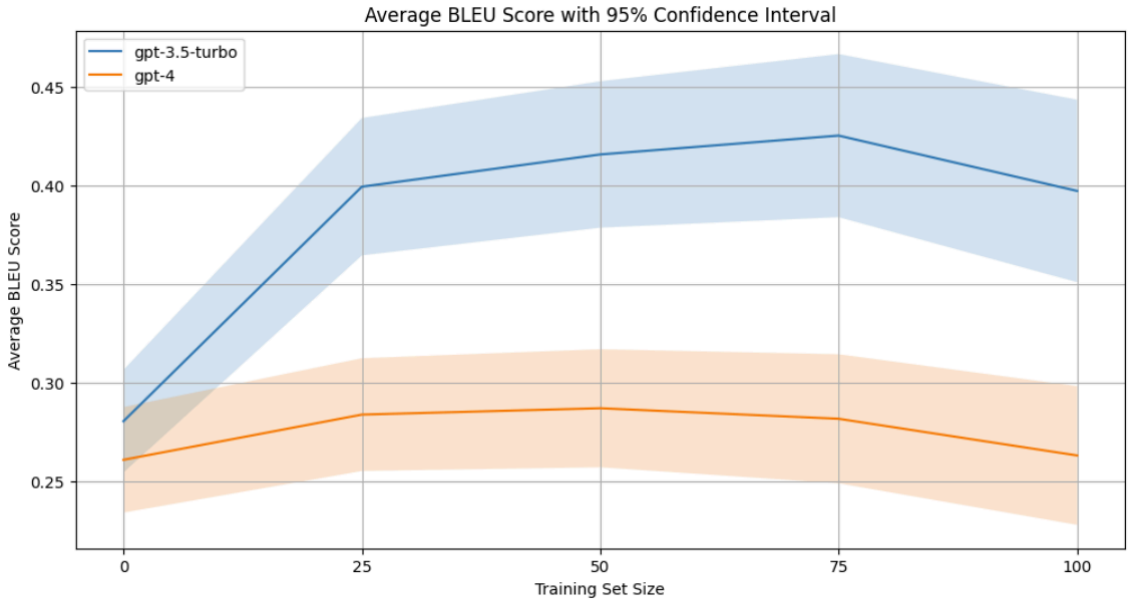
Overall, our analysis indicates the efficacy of using trained LLMs on clinically-relevant translation tasks. We found that a RAG-based 5-shot model was most accurate, and we believe RAG-based 5-shot is the best as it allows strong performance without requiring large volumes of data. This has significant real-world implications as the use of a popular LLM for a translation task would be significantly less accurate than even a slightly trained model. Many clinicians are still using Google Translate and Helsinki-NLP, and our model significantly outperforms both. We recommend that a staged rollout be implemented for clinicians, using a few-shot trained, secure LLM with human review. Limitations of the study include a small dataset, where a larger dataset may have led to performance gains with different k-shot learning variations and training test sizes.. Likewise, the only language included was Spanish. Lastly, the analysis only considered BLEU scores to compare model performance, which disregards potential gains in areas such as semantics. Future analysis should consider the use of a more comprehensive gold standard dataset, such as discharge summaries from different departments and in different languages, as well as potential human-in-the-loop augmentation to improve model performance on the translation task.

VI. APPENDIX

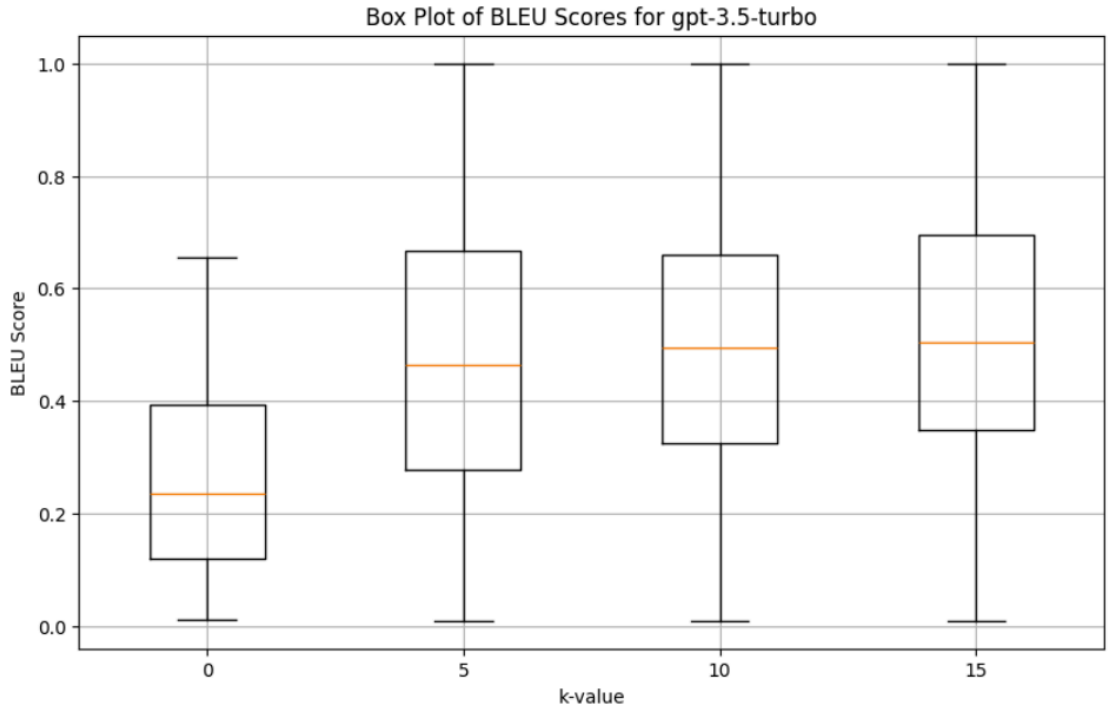
Appendix 5.0: Sample rows of patient case summaries with clinician-conducted Spanish translations. Patient case summaries were turned into machine readable format, parsed into 1 phrase sentences.

index	sentence_id	chart_id	sentence	Translation
0	1	6375371-5	an 18 year old female presented with repeated vomiting, headache, weakness of right upper limb and slurring of speech.	Una mujer de 18 años se presentó con vómitos recurrentes, cefalea, debilidad en el miembro superior derecho y disartria.
1	2	6375371-5	mri showed a left fronto parietal mass with midline shift.	La resonancia magnética (RM) mostró una masa en la región frontoparietal izquierda con desplazamiento de la línea media.
2	3	6375371-5	fronto parietal craniotomy and excision of the tumour were done with a pre operative diagnosis of high grade glioma.	Se realizó una craneotomía frontoparietal y la extirpación del tumor con un diagnóstico preoperatorio de glioma de alto grado.
3	4	6375371-5	histomorphologic examination and immunohistochemistry findings are summarised in .	Los hallazgos histomorfológicos y de inmunohistoquímica se resumen en .
4	5	6375371-5	a diagnosis of anaplastic pleomorphic xanthoastrocytoma was finally made (, and).\ntwo years post operatively, she came back with similar complaints and a repeat mass in the same site was found.	Finalmente, se estableció el diagnóstico de xantoastrocitoma pleomórfico anaplásico (, y).

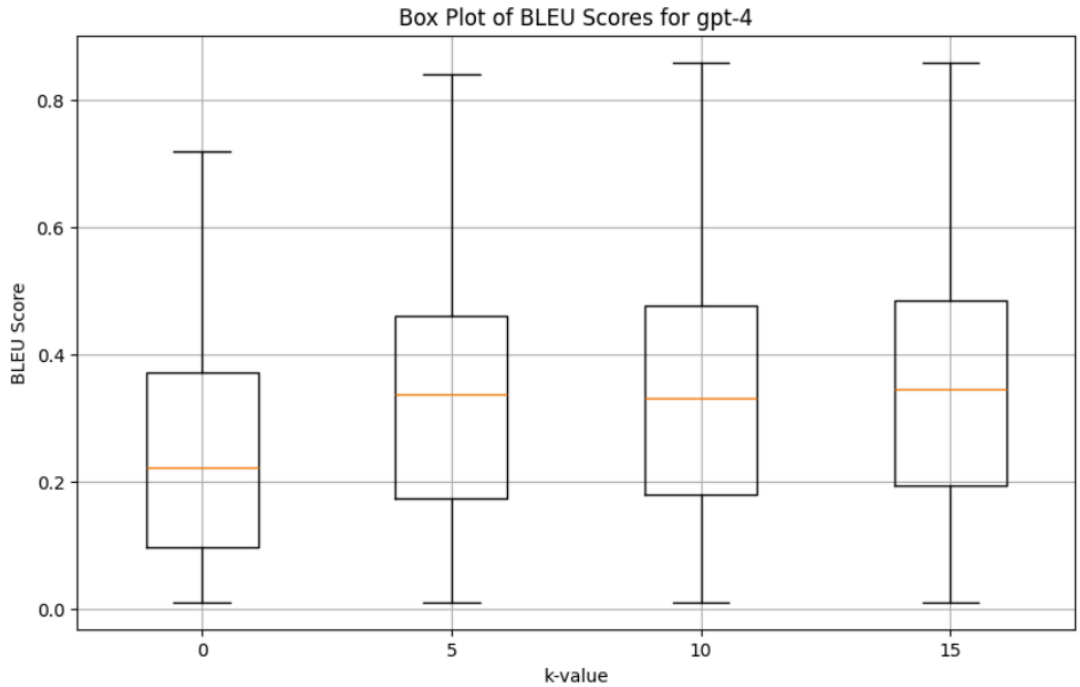
Appendix 5.1: Average BLEU scores calculated across k-values for random k-shot learning, compared between GPT-3.5-Turbo and GPT-4.



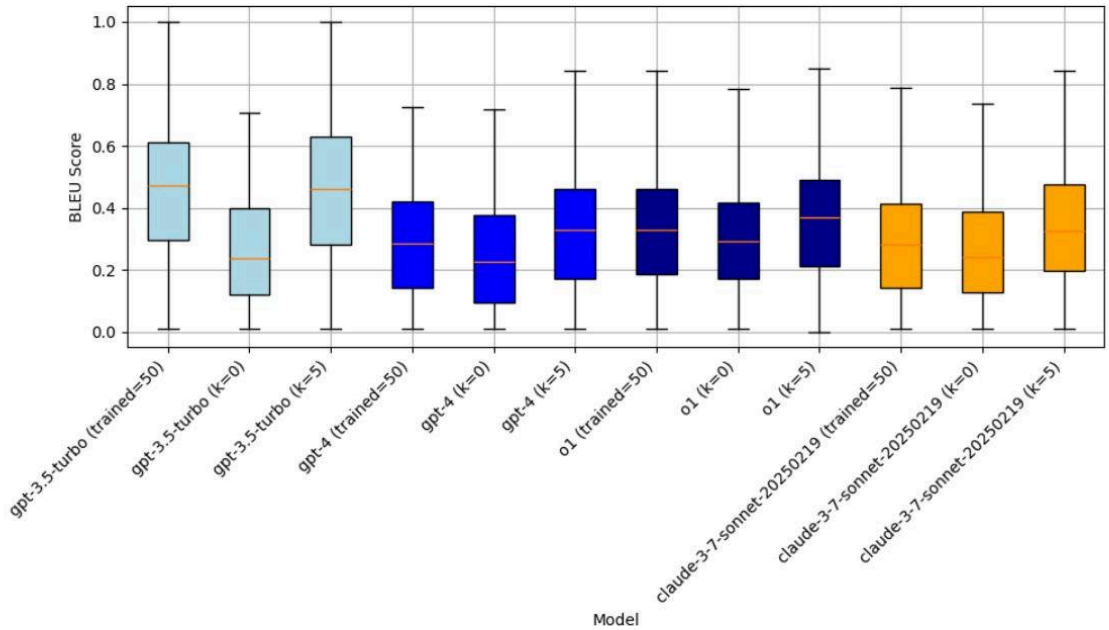
Appendix 5.2: Box plot of BLEU scores calculated across k-values for RAG-based k-shot learning for GPT-3.5-Turbo



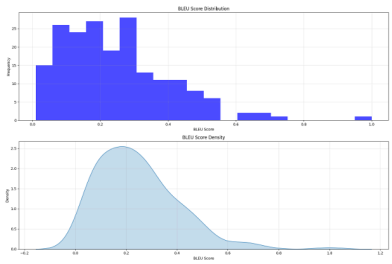
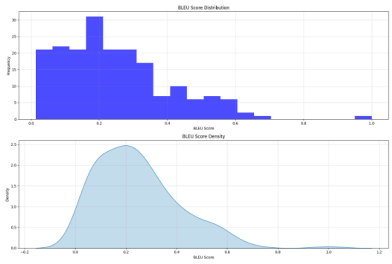
Appendix 5.3: Box plot of BLEU scores calculated across k-values for RAG-based k-shot learning for GPT4



Appendix 5.4: Box plot of BLEU scores across models LLMs and translation approaches with random K=50 (*denoted as trained =50*), K = 0 / untrained and RAG-based K = 5 few shot



Appendix 5.5: Comparison of Standard-of-Care Clinical Translation Models

	Mean BLEU Score (SD)	**BLEU score density**
Google Translate	0.2446 (0.158)	
Helsinki NLP	0.2457 (0.1628)